

VARAD KOTTAWAR

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EDUCATION

University of California, San Diego

Sep 2025 - Dec 2026 (Expected)

Master of Science in Computer Engineering, GPA: 3.7

- Coursework: Deep Learning Systems, Computer Architecture

Pune Institute of Computer Technology

Jul 2019 - Jun 2023

Bachelor of Engineering in Electronics and Telecommunication, GPA: 3.8

- Coursework: Data Structures and Algorithms, Object-Oriented Programming

EXPERIENCE

Z Lab, UC San Diego

Jul 2026 - Present

Researcher

- Deployed 4B-parameter LLM (Qwen3) to Qualcomm Snapdragon X Elite NPU using ONNX Runtime and QNN Execution Provider; implemented static-shape graph conversion, padded KV cache design, and INT8/INT16 post-training quantization pipeline with custom calibration data generation

Deutsche Bank

Jul 2023 - Aug 2025

Software Engineer

- Maintained 99% uptime for a backend serving 100,000+ employees by partnering with cross-functional teams to sustain 60,000+ monthly operations and a React frontend for 200+ leaders
- Boosted batch throughput 40% and enabled real-time search across 100,000+ employee records by building a low-latency Completion Suggester API using file-based transfer and in-memory caching
- Led a role-based access control system from ideation to launch with Spring Security for 1,000+ users, and shipped a real-time React simulator letting 200+ executives model org-restructuring scenarios

PUBLICATIONS

Recovering Internal Command Interfaces from Applications for Direct Agent Invocation

SOSP '26

Prince Modi, Varad Kottawar, Ayush Govind

PROJECTS

PagedQA

- Built a document Q&A system in Python leveraging vLLM's async engine with prefix caching, achieving 1.28x latency reduction, 91.2% KV cache hit rate, and 27% TTFT improvement at concurrency 10 on a T4 GPU
- Implemented a Gradio UI with real-time KV block memory visualization, streaming token generation, and an asyncio-based cold vs. warm benchmarking pipeline measuring latency, TTFT, and throughput across concurrent requests

GPU-Accelerated Profile HMM Viterbi Aligner

- Developed a GPU-accelerated sequence alignment system in CUDA using parallel dynamic programming on 32 SARS-CoV-2 genomic sequences, implementing on-device backtracking to eliminate host-device memory transfers and reduce pipeline latency
- Achieved 156x speedup over HMMER3 baseline by tuning CUDA thread block configurations, optimizing memory access patterns for coalesced reads and writes across parallel warp threads

CodeLens: Transformer-Powered RAG Code Review

- Built CodeLens, an end-to-end AI code-review system pairing a from-scratch PyTorch transformer with a ChromaDB RAG pipeline to deliver context-aware PR feedback via a 120B LLM
- Built a GitHub Actions CI pipeline that auto-fetches PR diffs and posts structured JSON reviews to PRs within 20s of open/update events

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, TypeScript, JavaScript, SQL

Frameworks: vLLM, PyTorch, TensorFlow, CUDA, Node.js, Flask, Spring Boot, React

Tools: AWS, Docker, MySQL, GitHub, OpenShift, CI/CD

AWARDS AND RECOGNITION

Winner, DataHacks 2026 (Best Use of Marimo/Sphinx): **Winner, DataHacks 2026** (Best Use of Marimo/Sphinx): Built Seismic Risk Atlas, an ML pipeline estimating household earthquake loss across 2,498 LA County census tracts

Awarded **Employee of the Quarter** for automating a high-frequency manual operation end to end, cutting repetitive engineering overhead and measurably improving team delivery speed